

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

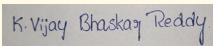
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO4 – Articulation (combining methods for evaluation) P4	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	Total Marks:	100
CO4 Statement:	Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)		
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Section / Batch:	AIML/2025	Date:	20/08/2026

Code of Conduct

I K.Vijay Bhaskar Reddy(192425155) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Industrial Inspection System – Comparison and Evaluation of Suitable Method

1. Introduction

An **Industrial Inspection System** is a system used in manufacturing industries to automatically inspect products, components, or materials and identify defects. Such systems commonly use **computer vision, image processing, machine learning, or deep learning** techniques to detect defects such as cracks, scratches, missing components, incorrect dimensions, surface damage, or manufacturing errors.

For an industrial inspection system to be practically useful, it must satisfy two important requirements:

1. **High accuracy:** The system must detect defects with an accuracy of **at least 95%**.
2. **Real-time operation:** The system must process and inspect products quickly enough to operate during the manufacturing process without significant delay.

Three possible methods are available:

- **Method A:** 92% accuracy and fast processing
- **Method B:** 96% accuracy and slow processing
- **Method C:** 95% accuracy and moderate processing speed

The three methods need to be compared based on accuracy, speed, engineering suitability, and the requirements of the industrial application.

2. Industry Problem Understanding

In an industrial environment, defective products must be identified before they reach the customer. Manual inspection can be time-consuming, expensive, and inconsistent because human inspectors may become tired or miss small defects.

An automated inspection system can continuously inspect products and provide consistent results. However, simply having high accuracy is not enough. The system must also be sufficiently fast to keep up with the production line.

For this problem, the minimum acceptable defect-detection accuracy is **95%**. Therefore:

$$\text{Accuracy} \geq 95\%$$

The system must also support **real-time operation**.

Therefore, the ideal method should have:

- Accuracy $\geq 95\%$
- Sufficient processing speed
- Reliable defect detection
- Ability to operate continuously
- Good balance between performance and computational requirements

3. Given Methods

Method	Accuracy	Processing Speed	Accuracy Requirement	Real-Time Requirement
Method A	92%	Fast	✗ Not satisfied	✓ Satisfied
Method B	96%	Slow	✓ Satisfied	✗ Not satisfied
Method C	95%	Moderate	✓ Satisfied	✓ Satisfied

4. Detailed Evaluation of Method A

Method A – 92% Accuracy, Fast

Method A provides a processing speed that is suitable for real-time applications. This is an important advantage because industrial production lines generally require fast inspection.

However, the accuracy of Method A is only:

92%

The required accuracy is:

95%

Therefore, the accuracy falls short by:

$95 - 92 = 3\%$

This means Method A does not satisfy the most important accuracy requirement.

Advantages

- Very fast processing
- Suitable for high-speed production lines
- Low processing delay
- Potentially lower computational requirements

Disadvantages

- Accuracy is only 92%
- Fails the minimum accuracy requirement
- May allow defective products to pass inspection
- May reduce product quality
- Not acceptable when 95% accuracy is mandatory

Evaluation

Although Method A is fast, **speed alone cannot compensate for insufficient accuracy**. In industrial inspection, missing defects can result in poor product quality, customer complaints, material wastage, and financial losses.

Therefore:

Method A is rejected because its accuracy is below the required 95%.

5. Detailed Evaluation of Method B

Method B – 96% Accuracy, Slow

Method B provides an accuracy of:

96%

This is higher than the required minimum of 95%.

The accuracy requirement is therefore satisfied.

However, Method B has **slow processing speed**. Real-time inspection requires the system to analyze products quickly while they are moving through the production line.

If the processing time is too high, the system may:

- Delay production
- Create a bottleneck
- Require products to wait for inspection
- Reduce production efficiency
- Fail to inspect every product in real time

Advantages

- Highest accuracy among the three methods
- Accuracy of 96% exceeds the requirement
- Better defect detection performance
- Suitable where accuracy is more important than speed

Disadvantages

- Slow processing
- Not suitable for real-time operation
- Can create production delays
- May require more computational resources
- Can reduce overall manufacturing throughput

Evaluation

Method B technically satisfies the accuracy requirement, but it fails the **real-time operation requirement**.

In this problem, both accuracy and real-time operation are required. Therefore, Method B cannot be selected despite having the highest accuracy.

Method B is rejected because its slow processing does not satisfy the real-time requirement.

6. Detailed Evaluation of Method C

Method C – 95% Accuracy, Moderate Speed

Method C provides an accuracy of exactly:

95%

The required accuracy is:

$\geq 95\%$

Therefore:

$95\% \geq 95\%$

Hence, Method C satisfies the accuracy requirement.

Method C also has **moderate processing speed**. Unlike Method B, it is not slow, and unlike Method A, it does not sacrifice the required accuracy for speed.

This makes Method C a balanced solution.

Advantages

- Meets the required 95% accuracy
- Provides moderate processing speed
- Suitable for real-time industrial inspection
- Provides a good balance between accuracy and speed
- Avoids the major limitations of Methods A and B
- Suitable for practical deployment

Disadvantages

- Accuracy is not higher than Method B
- Processing speed is not as fast as Method A
- It operates exactly at the minimum required accuracy, leaving little margin for performance degradation

Evaluation

Method C is the only method that satisfies **both major requirements simultaneously**.

Method C is therefore the most suitable method for the industrial inspection system.

7. Overall Comparison

Feature	Method A	Method B	Method C
Accuracy	92%	96%	95%
Minimum accuracy required	95%	95%	95%
Accuracy requirement	 Fail	 Pass	 Pass
Speed	Fast	Slow	Moderate
Real-time requirement	 Pass	 Fail	 Pass
Defect detection	Low	Very high	High
Production suitability	Low	Low	High
Overall decision	 Reject	 Reject	 Select

8. Decision Matrix

A simple decision matrix can also be used to evaluate the methods.

Let the two critical criteria be:

- **Accuracy**
- **Real-time speed**

Since both are essential requirements, a method that fails either requirement should not be selected.

Method	Accuracy ≥95%	Real-Time	Final Result
Method A			Rejected
Method B			Rejected
Method C			Selected

This clearly demonstrates that **Method C is the only method that satisfies both mandatory requirements.**

9. Engineering Concepts Applied

The selection of the method involves several important engineering concepts.

a) Accuracy

Accuracy represents how correctly the inspection system identifies defective and non-defective products.

A higher accuracy generally means fewer inspection errors.

The minimum acceptable accuracy is:

$$95\%$$

Method A fails because:

$$92\% < 95\%$$

Method B satisfies the requirement:

$$96\% > 95\%$$

Method C also satisfies it:

$$95\% = 95\%$$

b) Processing Speed

Processing speed is important because industrial production lines operate continuously.

A system with high accuracy but excessive processing time may not be useful in a high-speed manufacturing environment.

Therefore, the system must provide an appropriate trade-off between:

Accuracy and Speed

c) Real-Time Processing

Real-time processing means that the system can inspect products quickly enough without causing significant delays in the production process.

For this problem:

- Method A → Fast → suitable for real-time
- Method B → Slow → unsuitable for real-time
- Method C → Moderate → suitable for the stated requirement

d) Trade-off

Engineering design often involves a trade-off between competing performance parameters.

Here:

- Increasing speed may reduce accuracy.
- Increasing accuracy may increase processing time.

Method A demonstrates the **speed-over-accuracy** trade-off.

Method B demonstrates the **accuracy-over-speed** trade-off.

Method C provides a **balanced compromise**.

10. Solution Design

The proposed industrial inspection system using Method C can be designed as follows:

Step 1 – Image Acquisition

A camera captures images of products moving along the production line.

Step 2 – Pre-processing

The captured images can be processed using techniques such as:

- Noise removal
- Image resizing
- Grayscale conversion
- Contrast enhancement
- Filtering

Step 3 – Feature/Defect Analysis

The processed image is analyzed using an appropriate image-processing or machine-learning algorithm.

The system identifies characteristics such as:

- Cracks
- Scratches
- Holes
- Shape irregularities
- Missing components
- Surface defects

Step 4 – Classification

The product is classified as:

- **Defective**
- **Non-defective**

Step 5 – Decision

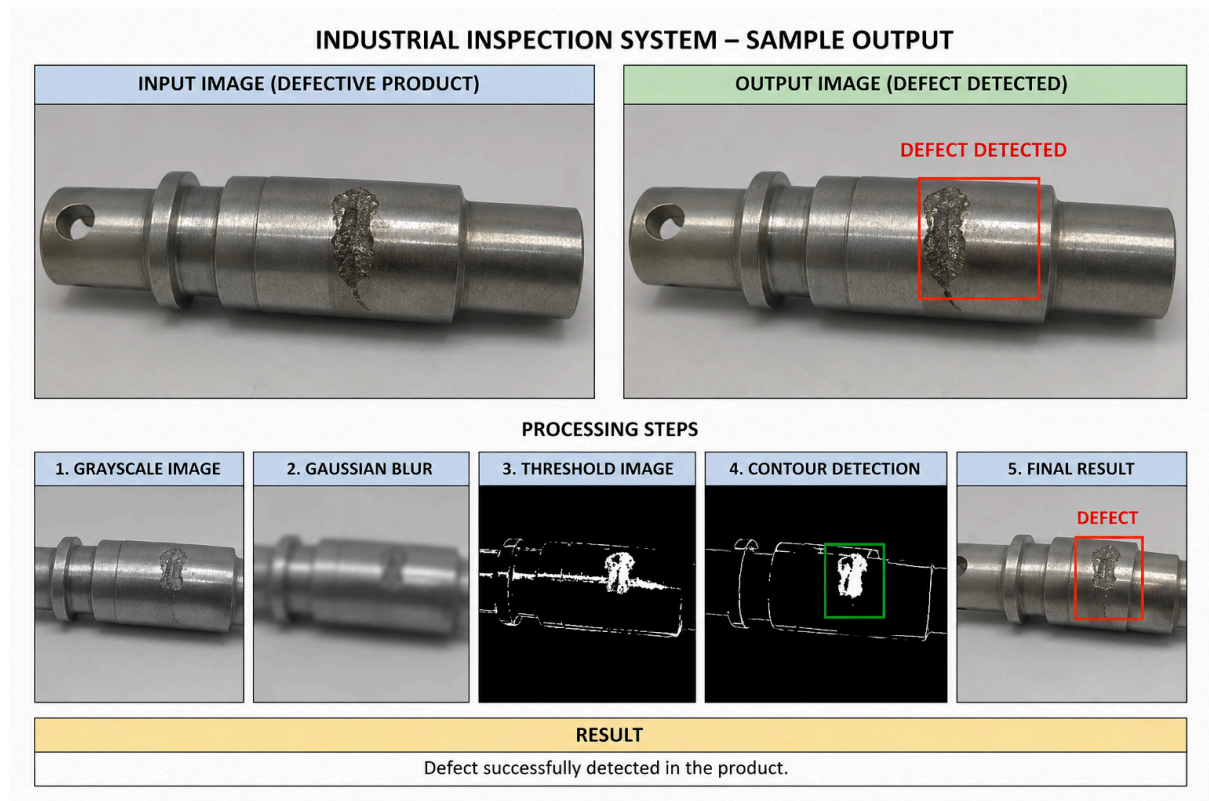
If a defect is detected, the system can send a signal to an actuator or controller to remove the defective product from the production line.

Step 6 – Continuous Operation

The system continuously repeats the inspection process for every product.

A simplified workflow is:

Product → Camera → Image Processing → Defect Detection → Classification → Decision → Reject/Accept



11. Use of Tools

An industrial inspection system can use various engineering and software tools.

Hardware

- Industrial camera
- Lighting system
- Computer or edge-processing device
- Conveyor belt
- Sensors
- Actuator/rejection mechanism

Software

Possible software tools include:

- Python
- OpenCV
- NumPy
- Machine-learning libraries
- Deep-learning frameworks

For example, **OpenCV** can be used for image acquisition and image processing, while a machine-learning model can be used for defect classification.

12. Performance Analysis

The three methods can be analyzed from both accuracy and speed perspectives.

Method A

Accuracy=92%Accuracy = 92\%

Although it is fast, it does not meet the minimum required accuracy.

Therefore:

Method A = Not Suitable\boxed{\text{Method A = Not Suitable}}

Method B

Accuracy=96%Accuracy = 96\%

It exceeds the accuracy requirement, but its slow speed prevents real-time operation.

Therefore:

Method B = Not Suitable\boxed{\text{Method B = Not Suitable}}

Method C

Accuracy=95%Accuracy = 95\%

It meets the minimum accuracy and provides moderate speed.

Therefore:

Method C = Suitable\boxed{\text{Method C = Suitable}}

13. Risk Analysis

Selecting an unsuitable method could create different risks.

Method A – Risk

The main risk is **false negatives**, where defective products are incorrectly classified as good.

Since its accuracy is only 92%, it does not meet the required standard.

Method B – Risk

The main risk is **processing delay**.

Even though it detects defects accurately, slow processing can interrupt the manufacturing workflow.

Method C – Risk

The primary risk is that it operates exactly at the minimum accuracy requirement. If real-world conditions reduce performance, accuracy could potentially fall below 95%.

To reduce this risk, the system should be properly calibrated and tested using representative industrial data.

14. Professional Evaluation

From an engineering perspective, selecting the method should not be based on a single performance parameter.

A professional evaluation considers:

1. Functional requirements
2. Accuracy
3. Processing speed
4. Reliability
5. Computational requirements
6. Production throughput
7. Cost
8. Scalability
9. Maintenance
10. Safety and quality requirements

In this particular problem, **accuracy $\geq 95\%$ and real-time operation are mandatory requirements**. Therefore, a method failing either requirement should be eliminated before considering secondary factors.

This makes the decision straightforward.

15. Final Decision

Method C is the recommended solution. 

The decision can be summarized as:

Method C: 95% Accuracy + Moderate Speed\boxed{\text{Method C: 95\% Accuracy + Moderate Speed}}

It satisfies both mandatory requirements:

Accuracy $\geq 95\%$ \boxed{Accuracy $\geq 95\%$ }

and

Real-Time Operation\boxed{\text{Real-Time Operation}}

Method A is rejected because:

92% < 95% 92\% < 95\%

Method B is rejected because its processing speed is too slow for real-time operation.

Method C provides the **best overall balance between defect-detection accuracy and processing speed**.

16. Conclusion

The industrial inspection system requires a method that can detect defects with a minimum accuracy of **95%** while also operating in **real time**.

Method A has excellent speed but only **92% accuracy**, so it fails the accuracy requirement.

Method B has the highest accuracy at **96%**, but its slow processing makes it unsuitable for real-time industrial applications.

Method C provides **95% accuracy** with **moderate processing speed**. It satisfies both the required accuracy and real-time operation requirements.

Therefore, **Method C is the most suitable method** for the industrial inspection system.